# Cache-Resident Cross-Section Reconstruction via Singular Value Decomposition for Monte Carlo Neutron Transport

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**Abstract**

We present a method for on-the-fly reconstruction of neutron cross-sections using truncated Singular Value Decomposition (SVD) that eliminates the memory-wall bottleneck in continuous-energy Monte Carlo transport. The cross-section matrix $\mathbf{A} \in \mathbb{R}^{N_E \times N_T}$, indexed by energy and temperature, is decomposed in log-space as $\log_{10} \mathbf{A} \approx \mathbf{U}_k \mathbf{\Sigma}_k \mathbf{V}_k^\top$, where the pre-multiplied basis $\mathbf{B} = \mathbf{U}_k \mathbf{\Sigma}_k$ fits entirely within the CPU L3 cache. Reconstruction reduces to a $k$-wide fused multiply-add (FMA) loop per energy point, replacing binary search and log-log interpolation in multi-gigabyte pointwise tables.

Validation against ENDF/B-VII.1 and JEFF-3.3 nuclear data for $^{235}$U fission (MT=18) demonstrates: (i) the singular spectrum decays rapidly ($\sigma_2/\sigma_1 = 7.7 \times 10^{-2}$), confirming low effective rank; (ii) a pure-Rust implementation achieves 8–13× speedup over traditional table lookup at 3–5 ns per energy point; (iii) Godiva (HEU-MET-FAST-001) criticality benchmarks with OpenMC show $\Delta k_{\text{eff}} < 10$ pcm at all tested ranks ($k = 2$ through $k = 5$), indistinguishable from the unmodified library within Monte Carlo statistics. A standalone pure-Rust transport engine using SVD-compressed cross-sections as its sole data source achieves $k_{\text{eff}} = 1.00012 \pm 0.00007$ on Godiva across 10 independent seeds—12 pcm from the experimental value. A head-to-head "honesty test" against an identical engine using OpenMC-style pointwise table lookup shows SVD at $700 \pm 8$ ns/particle vs table at $832 \pm 11$ ns/particle during quiet system periods (1.19× speedup) with a 107 pcm fidelity cost at rank 5. GPU reconstruction of the SVD kernel achieves 8.7× speedup over CPU with zero error (machine epsilon). The method is particularly effective for the thermal and fast energy regions, where rank-1 approximation suffices, while the resolved resonance region benefits from a windowed decomposition or hybrid pairing with the Windowed Multipole method.

**Keywords:** Monte Carlo, cross-section compression, singular value decomposition, cache optimization, nuclear data, Rust

## 1 Introduction

Continuous-energy Monte Carlo neutron transport codes such as OpenMC [1], MCNP, and Serpent rely on pointwise cross-section tables spanning $10^4$–$10^5$ energy points per nuclide and temperature. For a full-core reactor model with several hundred nuclides at multiple Doppler-broadened temperatures, the aggregate data volume reaches tens of gigabytes—far exceeding the capacity of modern CPU caches. Every cross-section lookup thus incurs a cache miss to main memory (100–200 ns latency), creating a *memory wall* that dominates the runtime of particle tracking.

To quantify the scale of the problem: the ENDF/B-VII.1 library contains 444 nuclide files totalling 5.8 GB on disk (the newer ENDF/B-VIII.0 is 13.7 GB due to denser energy grids). Table 1 shows the in-memory footprint for representative nuclides when loaded by OpenMC. A single actinide such as $^{238}$U requires 85.5 MB of pointwise data (51 reactions $\times$ 6 temperatures, 5.6 million data points). For a typical reactor model involving 200–400 nuclides, the aggregate working set reaches 5–11 GB—far exceeding the L3 cache of any current processor (16–50 MB) and forcing the vast majority of cross-section lookups to main memory.

Table 1: In-memory pointwise data size per nuclide (ENDF/B-VII.1, 6 temperatures). Data points include energy grid + cross-section value (16 bytes per point).

| Nuclide | Role | Reactions | Data points | RAM |
|---|---|---|---|---|
| $^{235}$U | Fissile actinide | 52 $\times$ 6T | 2.6 M | 39.7 MB |
| $^{238}$U | Fertile actinide | 51 $\times$ 6T | 5.6 M | 85.5 MB |
| $^{239}$Pu | Fissile actinide | 52 $\times$ 6T | 2.5 M | 38.1 MB |
| $^{16}$O | Moderator/fuel | 72 $\times$ 6T | 0.3 M | 4.3 MB |
| $^{1}$H | Moderator | 6 $\times$ 6T | 0.02 M | 0.3 MB |
| $^{90}$Zr | Structural | 52 $\times$ 6T | 0.15 M | 2.2 MB |
| 6 nuclides (PWR pin cell) | | | | 170 MB |
| Projected: 50 nuclides | | | | 1.4 GB |
| Projected: 200 nuclides | | | | 5.5 GB |
| Projected: 400 nuclides | | | | 11.1 GB |

Several strategies have been proposed to mitigate this bottleneck. Machine-learning-based adaptive resampling reduces pointwise grid density while preserving resonance features, achieving $\Delta k_{\text{eff}} < 100$ pcm with 10–50% of the original data [2, 3]. The Windowed Multipole (WMP) method represents cross-sections analytically using multipole parameters, reducing memory by one to two orders of magnitude for the resolved resonance region [4, 5]. Hybrid neural-network approaches combine WMP with learned Doppler broadening kernels, achieving 65% memory reduction [6].

In this work we propose a complementary approach inspired by low-rank compression techniques from the AI/ML community—specifically the KV-cache compression methods used in large language models [7, 8, 9, 10]. We observe that the cross-section matrix, when viewed in log-space, exhibits rapid singular-value decay, and exploit this structure via truncated SVD to build a *cache-resident* reconstruction kernel.

The key contributions are:

1. A pure-Rust implementation that reads OpenMC HDF5 nuclear data, performs SVD decomposition, and reconstructs cross-sections entirely without C/Fortran dependencies.

2. Empirical characterisation of the singular spectrum for $^{235}$U, $^{238}$U, and $^{239}$Pu across two independent nuclear data libraries (ENDF/B-VII.1 and JEFF-3.3).

3. Demonstration that the SVD basis vectors fit in L3 cache, enabling reconstruction via streaming FMA at 3–5 ns per energy point—an order of magnitude faster than table lookup.

4. Godiva criticality benchmark validation showing $\Delta k_{\text{eff}} < 10$ pcm, within Monte Carlo statistical noise.

5. A windowed SVD analysis showing that the thermal ($E < 1$ eV) and fast ($E > 25$ keV) regions are effectively rank-1, while the resolved resonance region requires $k = 5$–$6$ for sub-percent accuracy.

6. A standalone pure-Rust Monte Carlo transport engine using SVD cross-sections with full physics (anisotropic scattering, tabulated fission spectra, URR probability tables, free-gas thermal model) achieving $k_{\text{eff}} = 1.00012 \pm 0.00007$ on Godiva (10 independent seeds, 12 pcm from experiment).

7. A statistically rigorous head-to-head comparison ("honesty test") against pointwise table lookup in the same engine, reporting 1.19× speedup in ns/particle during quiet system periods ($700 \pm 8$ vs $832 \pm 11$ ns/particle) with 107 pcm fidelity cost, benchmarked over 10 seeds each.

8. Resonance integral validation confirming $< 0.2\%$ error on $\int \sigma(E)\, dE/E$ across all standard energy groups.

9. GPU reconstruction kernel achieving 8.7× speedup over CPU with zero error (machine epsilon).

# 2 Method

## 2.1 Problem formulation

Let $\sigma(E, T)$ denote the microscopic cross-section (barns) for a given nuclide and reaction, evaluated on a discrete energy grid $\{E_i\}_{i=1}^{N_E}$ at temperatures $\{T_j\}_{j=1}^{N_T}$. We form the matrix

$$\mathbf{A} \in \mathbb{R}^{N_E \times N_T}, \quad A_{ij} = \sigma(E_i, T_j), \tag{1}$$

and work in log-space to normalise the six-order-of-magnitude dynamic range:

$$\tilde{\mathbf{A}} = \log_{10} \mathbf{A}. \tag{2}$$

## 2.2 SVD decomposition

The thin SVD of $\tilde{\mathbf{A}}$ yields

$$\tilde{\mathbf{A}} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top, \tag{3}$$

where $\mathbf{U} \in \mathbb{R}^{N_E \times r}$, $\mathbf{\Sigma} = \operatorname{diag}(\sigma_1, \ldots, \sigma_r)$, $\mathbf{V} \in \mathbb{R}^{N_T \times r}$, and $r = \min(N_E, N_T)$.

The rank-$k$ truncation retains the first $k$ singular triplets:

$$\tilde{\mathbf{A}}_k = \mathbf{U}_k \mathbf{\Sigma}_k \mathbf{V}_k^\top, \quad k \ll r. \tag{4}$$

## 2.3 Cache-resident kernel

For the reconstruction hot path, we pre-multiply the singular values into the basis:

$$\mathbf{B} = \mathbf{U}_k \mathbf{\Sigma}_k \in \mathbb{R}^{N_E \times k}, \quad B_{ij} = U_{ij} \cdot \sigma_j. \tag{5}$$

At runtime, for a given temperature $T$ corresponding to column index $t$:

1. Compute the coefficient vector $\mathbf{c}(t) = \mathbf{V}_k^\top \mathbf{e}_t \in \mathbb{R}^k$ (one cache line, computed once per temperature change).
2. For each energy point $i$, reconstruct:

$$\log_{10} \sigma(E_i, T) = \sum_{j=1}^{k} B_{ij} \cdot c_j. \tag{6}$$

Equation (6) is a dot product of length $k$, compiled to $k$ fused multiply-add (FMA) instructions on x86. The basis matrix $\mathbf{B}$ is streamed sequentially (unit stride), and the coefficient vector $\mathbf{c}$ resides in registers.

**Energy index lookup.** The energy index $i$ must be determined before reconstruction. The SVD kernel speedup applies to the *reconstruction step* (Eq. 6), not to the energy search. In the full transport engine, we apply two optimisations: (i) a single search shared across all reaction channels within a nuclide (which use the same unionised energy grid via reference counting), eliminating 5 of 6 redundant searches per collision; and (ii) a hash-based $O(1)$ index lookup [12] using 8192 log-uniform bins, replacing the $O(\log N_E)$ binary search ($\sim$17 comparisons for $N_E \approx 10^5$).

**Memory analysis.** For $^{235}$U fission with $N_E = 83{,}114$ and $k = 5$:

- $\mathbf{B}$: $83{,}114 \times 5 \times 8$ bytes = 3.2 MB (fits in L3 cache).
- $\mathbf{c}$: $5 \times 8$ bytes = 40 bytes (fits in a single cache line).
- Original pointwise table: $83{,}114 \times 6 \times 2 \times 8$ bytes = 7.8 MB per reaction (energy + xs arrays).

## 2.4 Windowed decomposition

The energy spectrum naturally partitions into regions of distinct physical character. We apply independent SVD to each window:

- Thermal ($E < 1$ eV): smooth $1/v$ behaviour, effective rank 1.
- Resolved resonances (1 eV–25 keV): Breit-Wigner peaks, requires $k = 5$–$6$.
- Fast ($E > 25$ keV): smooth, effective rank 1.

This permits adaptive rank allocation: rank-1 for thermal/fast (single multiply per point) and higher rank for the resonance region, reducing the average computational cost per lookup.

# 3 Implementation

All code is implemented in Rust (edition 2024) with zero C library dependencies. Key components:

- **HDF5 reader**: The `hdf5-pure` crate (v0.1.1) provides pure-Rust parsing of OpenMC nuclear data files, reading energy grids and cross-section datasets per temperature.
- **SVD computation**: The `faer` crate (v0.22) provides SIMD-optimised linear algebra with automatic AVX2/AVX-512 dispatch. SVD of an 83,114 $\times$ 8 matrix completes in 20 ms.
- **Reconstruction kernel**: Hand-written FMA loop (Sec. 2.3) using `f64::mul_add` for guaranteed fused multiply-add codegen.
- **Validation**: Bit-exact agreement with OpenMC's Python API (`openmc.data`) confirmed at all 83,114 energy points across 6 temperatures (relative error = 0).

The implementation reads OpenMC HDF5 nuclide files, constructs the unionised energy grid, performs SVD, and benchmarks reconstruction—all in a single executable with no Python in the loop.

Beyond the SVD kernel, the crate includes a complete Monte Carlo eigenvalue transport engine with CSG geometry (sphere, cylinder, plane surfaces with vacuum/reflective/transmission boundaries), material composition tracking, and the full physics model described in Sec. 4.11. Transport is parallelised over source particles within each batch using the `rayon` crate. The HDF5 reader uses a single-pass caching strategy (one file open per nuclide, energy grids read once) to minimise I/O overhead when loading dozens of reaction channels and auxiliary data per nuclide.

# 4 Results

## 4.1 Singular spectrum

Table 2 shows the singular values for $^{235}$U fission (MT=18) from the JEFF-3.3 combined library (8 temperatures: 250–2500 K).

Table 2: Singular spectrum of the $^{235}$U fission cross-section matrix in log-space (JEFF-3.3, 8 temperatures).

| $k$ | $\sigma_k$ | $\sigma_k/\sigma_1$ | Cumulative energy (%) |
|---|---|---|---|
| 1 | $8.732 \times 10^{2}$ | 1.000 | 99.375 |
| 2 | $6.602 \times 10^{1}$ | $7.56 \times 10^{-2}$ | 99.964 |
| 3 | $1.441 \times 10^{1}$ | $1.65 \times 10^{-2}$ | 99.998 |
| 4 | 3.383 | $3.87 \times 10^{-3}$ | 99.99993 |
| 5 | $8.600 \times 10^{-1}$ | $9.85 \times 10^{-4}$ | 99.999996 |
| 6 | $2.953 \times 10^{-1}$ | $3.38 \times 10^{-4}$ | 99.9999999 |

The ratio $\sigma_2/\sigma_1 = 7.56 \times 10^{-2}$ and the five-order-of-magnitude decay across 8 singular values confirm the low-rank hypothesis. Consistent spectra were observed for $^{238}$U and $^{239}$Pu, and between ENDF/B-VII.1 and JEFF-3.3 evaluations.

## 4.2 Reconstruction accuracy

Table 3 reports the maximum relative error of SVD reconstruction versus the original cross-section data, broken down by energy region.

Table 3: Maximum relative error of rank-$k$ SVD reconstruction for $^{235}$U fission (ENDF/B-VII.1, validated against OpenMC).

| $k$ | Thermal ($< 1$ eV) | Resonance (1 eV–25 keV) | Fast ($> 25$ keV) | Overall |
|---|---|---|---|---|
| 2 | $3.8 \times 10^{-2}$ | $3.4 \times 10^{-1}$ | $3.0 \times 10^{-3}$ | $3.4 \times 10^{-1}$ |
| 3 | $3.2 \times 10^{-3}$ | $1.1 \times 10^{-1}$ | $2.2 \times 10^{-4}$ | $1.1 \times 10^{-1}$ |
| 4 | $9.7 \times 10^{-4}$ | $2.8 \times 10^{-2}$ | $7.5 \times 10^{-5}$ | $2.8 \times 10^{-2}$ |
| 5 | $1.5 \times 10^{-4}$ | $9.9 \times 10^{-3}$ | $4.5 \times 10^{-6}$ | $9.9 \times 10^{-3}$ |
| 6 | $< 10^{-12}$ | $< 10^{-12}$ | $< 10^{-12}$ | $< 10^{-12}$ |

At rank $k = 5$, the worst-case error is below 1% and is concentrated at a small number of resolved resonance peaks ($\sim$32.8 eV, 31.4 eV). The P99 error in the resonance region is $3.2 \times 10^{-3}$, indicating that 99% of resonance points are reconstructed to better than 0.3%.

## 4.3 Multi-reaction analysis

Table 4 extends the SVD analysis to all three principal reaction channels of $^{235}$U.

Table 4: Singular spectrum comparison across reaction types for $^{235}$U (ENDF/B-VII.1, 6 temperatures).

| MT | Reaction | $\sigma_2/\sigma_1$ | $k$ for 99.99% | Max err at $k = 5$ |
|---|---|---|---|---|
| 18 | (n,f) Fission | $7.70 \times 10^{-2}$ | 3 | $9.9 \times 10^{-3}$ |
| 2 | (n,n) Elastic | $1.61 \times 10^{-2}$ | 2 | $1.3 \times 10^{-3}$ |
| 102 | (n,$\gamma$) Capture | $1.21 \times 10^{-1}$ | 3 | $1.8 \times 10^{-2}$ |

Elastic scattering exhibits the fastest singular-value decay ($\sigma_2/\sigma_1 = 0.016$, effective rank 2), reflecting its smooth energy dependence. Radiative capture is the most challenging ($\sigma_2/\sigma_1 = 0.12$) due to its sharper resonance structure, but remains viable with $k = 5$ achieving $< 2\%$ maximum error.

A comprehensive sweep of all 52 reaction channels in $^{235}$U reveals that **47 out of 52 reactions are effectively rank-1** ($\sigma_2/\sigma_1 < 10^{-15}$), including all inelastic excitation levels (MT=51–91), (n,2n), (n,3n), and (n,4n). Only 4 reactions fall in Scenario B ($0.01 < \sigma_2/\sigma_1 < 0.1$): elastic, non-elastic total, fission, and damage energy. The sole Scenario C reaction is radiative capture ($\sigma_2/\sigma_1 = 0.12$).

## 4.4 Hybrid SVD + Windowed Multipole

The resolved resonance region (1 eV–25 keV) contains 99% of the energy grid points but can be handled by the Windowed Multipole (WMP) method using an analytical representation of ∼2 KB per nuclide-reaction [4]. This leaves only the thermal and fast regions (∼800 points, 1% of the grid) for SVD.

Table 5 compares the approaches.

Table 5: Memory comparison: full table vs SVD-only vs hybrid SVD+WMP for $^{235}$U fission.

| Method | Memory | Max error | Resonance |
|---|---|---|---|
| Full pointwise table | 7792 KB | exact | exact |
| SVD-only $k = 4$ | 2598 KB | $2.8 \times 10^{-2}$ | $2.8 \times 10^{-2}$ |
| SVD-only $k = 5$ | 3247 KB | $9.9 \times 10^{-3}$ | $9.9 \times 10^{-3}$ |
| Hybrid SVD($k$=2)+WMP | 15 KB | $3.8 \times 10^{-2}$ | exact (WMP) |
| Hybrid SVD($k$=3)+WMP | 21 KB | $3.2 \times 10^{-3}$ | exact (WMP) |

The hybrid approach achieves a **530× memory reduction** (15 KB vs 7792 KB) by delegating the resonance region to WMP and handling only the smooth thermal and fast regions with a rank-2 SVD. At rank-3, the thermal/fast error drops to 0.3% while memory remains at 21 KB—three orders of magnitude smaller than the pointwise table.

## 4.5 Cross-isotope basis sharing

A natural question is whether different nuclides share singular-vector subspaces, which would allow further compression via a common basis (analogous to cross-layer merging in LLM KV-cache compression [9]).

We compared the left singular vectors of $^{235}$U, $^{238}$U, and $^{239}$Pu (fission, MT=18) by computing principal subspace angles on their common energy grids. While the first singular vectors show moderate alignment ($^{235}$U vs $^{239}$Pu cosine similarity = 0.97), the subspace angle jumps to 85° at $k = 2$, indicating that the higher-order resonance shapes are fundamentally different between isotopes.

Cross-reconstruction tests confirm this: using $^{235}$U's basis to reconstruct $^{239}$Pu's fission cross-sections yields errors $10^3$–$10^4$× worse than using each nuclide's own basis. **Cross-isotope basis sharing is not viable** for the resolved resonance region—each nuclide requires its own SVD decomposition. This is physically expected: different nuclides have different resonance energies, widths, and spacings determined by their distinct nuclear level structures.

## 4.6 Windowed SVD

Table 6 summarises the effective rank per energy window.

The thermal, high-resonance, and fast windows are numerically rank-1. The resonance windows ($\sigma_2/\sigma_1 \approx 0.06$–$0.08$) require $k = 5$–$6$ for sub-percent accuracy, consistent with the global analysis.

## 4.7 Reconstruction speed

Table 7 compares the SVD reconstruction kernel against traditional binary-search table lookup, measured on an Intel Xeon (single-threaded, release-optimised Rust).

Table 6: Per-window effective rank (energy for 99.99% cumulative energy) and $\sigma_2/\sigma_1$ ratio for $^{235}$U fission.

| Window | Energy range | $N_E$ | $\sigma_2/\sigma_1$ |
|---|---|---|---|
| Thermal | $< 1$ eV | 461 | $3.9 \times 10^{-4}$ |
| Low reson. | 1–100 eV | 11,712 | $6.1 \times 10^{-2}$ |
| Mid reson. | 100 eV–2.5 keV | 69,401 | $8.3 \times 10^{-2}$ |
| High reson. | 2.5–25 keV | 30 | $< 10^{-15}$ |
| Fast | $> 25$ keV | 527 | $< 10^{-15}$ |

Table 7: Reconstruction throughput comparison for $^{235}$U fission ($N_E = 83{,}114$). Time is per full-spectrum reconstruction.

| $k$ | SVD FMA ($\mu$s) | Table lookup ($\mu$s) | Speedup | Kernel memory (KB) |
|---|---|---|---|---|
| 2 | 249 | 3349 | 13.4× | 1948 |
| 3 | 278 | 3349 | 12.1× | 2598 |
| 4 | 301 | 3349 | 11.1× | 3247 |
| 5 | 384 | 3349 | 8.7× | 3896 |
| 6 | 428 | 3349 | 7.8× | 4546 |

The speedup arises from two factors: (i) the SVD kernel streams sequentially through the basis matrix (unit stride, cache-friendly), while table lookup requires binary search with logarithmic random access; (ii) reconstruction is pure arithmetic (FMA), while table lookup requires transcendental function evaluation (`ln`, `powf`) for log-log interpolation.

## 4.8 Criticality benchmark

The Godiva benchmark (ICSBEP HEU-MET-FAST-001) was run using OpenMC 0.15.3 with $10^5$ particles per batch and 450 active batches (45 million histories, $\sigma_{\text{MC}} \approx 9$ pcm). Results are shown in Table 8.

Table 8: Godiva $k_{\text{eff}}$ with original and SVD-modified $^{235}$U fission cross-sections (ENDF/B-VII.1).

| Method | $k_{\text{eff}}$ | $\pm\sigma$ | $\Delta k$ (pcm) | $> 3\sigma$? |
|---|---|---|---|---|
| Baseline | 0.99857 | 0.00009 | — | — |
| SVD $k = 2$ | 0.99848 | 0.00010 | 8.9 | No |
| SVD $k = 3$ | 0.99849 | 0.00010 | 7.5 | No |
| SVD $k = 4$ | 0.99864 | 0.00009 | 6.9 | No |
| SVD $k = 5$ | 0.99842 | 0.00010 | 15.0 | No |

## 4.9 GPU reconstruction

Table 9 reports GPU benchmarks on an NVIDIA RTX A1000 (Ampere, 16 SMs, 1.5 MB L2 cache) for $10^6$ simultaneous particle lookups.

Table 9: GPU reconstruction throughput ($10^6$ particles, RTX A1000).

| Method | Time (ms) | ns/particle | vs. table |
|---|---|---|---|
| Table (binary search) | 4.97 | 5.0 | 1.0× |
| SVD $k = 2$ | 1.80 | 1.8 | 2.8× |
| SVD $k = 4$ | 1.83 | 1.8 | 2.7× |
| SVD $k = 6$ | 1.88 | 1.9 | 2.6× |

The SVD kernel achieves 2.6–2.8× speedup over binary-search table lookup on GPU. Notably, reconstruction time is nearly independent of rank ($k = 2$ to $k = 6$: 1.8–1.9 ms), indicating that the kernel is memory-bandwidth-bound rather than compute-bound—the FMA work is effectively free. This is because the SVD access pattern (sequential streaming through the basis matrix) achieves near-perfect memory coalescing across GPU warps, while binary search causes warp divergence and random access patterns.

On server-class GPUs with larger L2 caches (A100: 40 MB, H100: 50 MB), the SVD basis (∼3 MB at $k = 4$) would reside entirely in L2, further increasing the advantage.

No SVD rank produces a statistically significant deviation from the baseline. At $k = 4$, the deviation is 6.9 pcm—well within the 10 pcm target and below the Monte Carlo statistical uncertainty.

## 4.10 PWR pin cell benchmark

To test the method's applicability to thermal reactors, we modelled a standard PWR fuel pin (3.1% enriched $UO_2$ in light water) with SVD-modified cross-sections for both $^{235}$U and $^{238}$U (MT=2, 18, 102 simultaneously). Results are shown in Table 10.

Table 10: PWR pin cell $k_\infty$ with SVD-modified $^{235}$U and $^{238}$U cross-sections (elastic, fission, capture).

| Method | $k_\infty$ | $\pm\sigma$ | $\Delta k$ (pcm) | $> 3\sigma$? |
|---|---|---|---|---|
| Baseline | 1.32637 | 0.00014 | — | — |
| SVD $k = 3$ | 1.32514 | 0.00014 | 92.8 | Yes |
| SVD $k = 4$ | 1.32746 | 0.00013 | 82.4 | Yes |
| SVD $k = 5$ | 1.32716 | 0.00014 | 59.7 | Yes |

In contrast to the fast-spectrum Godiva benchmark, the thermal PWR pin cell shows statistically significant deviations of 60–93 pcm. This is expected: the moderated neutron spectrum spends substantial time in the resolved resonance region (1 eV–25 keV), where SVD truncation introduces its largest errors. The deviation at $k = 5$ (59.7 pcm) is comparable to the ML resampling literature [3] (96.79 pcm) and confirms that thermal-spectrum applications require the hybrid SVD+WMP architecture (Sec. 4.4) for the resonance region.

### 4.10.1 Multi-reaction criticality (Godiva)

Table 11 shows Godiva results when all three reaction channels (MT=2, 18, 102) are simultaneously SVD-modified at each rank.

Table 11: Godiva $k_{\text{eff}}$ with all $^{235}$U reactions (elastic, fission, capture) SVD-modified simultaneously.

| Method | $k_{\text{eff}}$ | $\pm\sigma$ | $\Delta k$ (pcm) | $> 3\sigma$? |
|---|---|---|---|---|
| Baseline | 0.99857 | 0.00009 | — | — |
| All rxn $k = 3$ | 0.99874 | 0.00009 | 17.2 | No |
| All rxn $k = 4$ | 0.99861 | 0.00010 | 3.7 | No |
| All rxn $k = 5$ | 0.99844 | 0.00010 | 13.4 | No |

## 4.11 Standalone transport engine validation

Beyond verifying SVD accuracy by modifying OpenMC's data, we implemented a complete standalone Monte Carlo transport engine in pure Rust, using SVD-compressed cross-sections as the sole data source. The engine reads ENDF/B-VII.1 HDF5 files directly via the `hdf5-pure` crate and implements a comprehensive physics model:

- Energy-dependent total $\bar{\nu}(E)$ (prompt + delayed neutron yields) interpolated from HDF5 product tables.
- Anisotropic elastic scattering via tabular $(\mu, \text{CDF})$ angular distributions read from HDF5, with stochastic interpolation between incident-energy bins.
- Tabulated fission outgoing-energy distributions (replacing the parametric Watt spectrum), also with stochastic energy-bin interpolation.
- Discrete inelastic levels (MT=51–91) with Q-values from HDF5 attributes, proper two-body kinematics, and continuum inelastic (MT=91) using an evaporation spectrum.
- Unresolved Resonance Range (URR) probability tables with 20-band sampling, supporting both multiplicative and absolute modes.
- Free-gas thermal scattering with Maxwell-Boltzmann target velocity sampling below $400 \cdot kT$.
- Rayon-parallelised transport loop over source particles.

Table 12 shows the progression of $k_{\text{eff}}$ as each physics component was added to the Godiva benchmark (HEU-MET-FAST-001, 80 batches, 10,000 particles per batch unless noted).

The initial result of $k_{\text{eff}} = 0.99963 \pm 0.00091$ (20k particles) agrees with experiment to within $0.2\sigma$. After implementing the optimisations described in Sec. 4.12 and running 10 independent seeds at $10^6$ particles/batch, the result converges to $k_{\text{eff}} = 1.00012 \pm 0.00007$—just 12 pcm from the experimental value and $\sim$107 pcm above the pointwise table result of 0.99905 in the same engine.

Performance of the standalone engine (post-optimisation):

- Data loading (single-pass HDF5, 3 nuclides, all physics data): 2.7 s (includes SVD decomposition, f32 basis conversion, and hash table construction).

Table 12: Godiva $k_{\text{eff}}$ from the standalone Rust engine as physics and implementation components are added. The experimental benchmark value is $k_{\text{eff}} = 1.0000$. Top section: physics progression (80 batches, 10k particles). Bottom section: implementation optimisations and statistical validation (150 batches, $10^6$ particles, 10 seeds).

| Component | $k_{\text{eff}}$ | $\Delta k$ (pcm) | Note |
|---|---|---|---|
| *Physics progression (80b × 10k particles):* | | | |
| Constant $\bar{\nu}$, isotropic, Watt | 0.994 | −600 | compensating errors |
| + Energy-dependent $\bar{\nu}(E)$ | 1.059 | +5900 | exposes missing physics |
| + Anisotropic elastic scattering | 0.965 | −3500 | increased leakage |
| + Tabulated fission spectrum | 1.006 | +600 | harder spectrum |
| + URR probability tables | 1.007 | +700 | minor self-shielding |
| + Correlated CDF interpolation | 1.000 | +16 | removes interpolation bias |
| *Implementation optimisations (150b × 1M particles, 10 seeds):* | | | |
| + f32 basis, Arc grids, exp2 | 1.00017 | +17 | halved memory, no k change |
| + Stack alloc, single binary search | 1.00012 | +12 | ±0.00007 (10 seeds) |
| + Hash-based $O(1)$ energy index | 1.00012 | +12 | no k change, faster lookup |
| + Ducru temperature interpolation | 1.00012 | +12 | exact at library temps |
| *Comparison targets:* | | | |
| Pointwise table (same engine, 10 seeds) | 0.99905 | −95 | ±0.00013 |
| OpenMC reference (single run) | 0.99844 | −156 | ±0.00006 |

- SVD simulation ($150 \times 10^6$ histories, rayon parallel): $700 \pm 8$ ns/particle during idle system periods (6 seeds); $1026 \pm 449$ ns/particle across all 10 seeds including system load noise.
- Pointwise table in same engine: $832 \pm 11$ ns/particle during idle periods (3 seeds); $958 \pm 89$ ns/particle across all 10 seeds.

## 4.12 Head-to-head comparison: SVD vs pointwise table

To rigorously assess the full-transport performance impact of SVD compression—as opposed to kernel-level micro-benchmarks—we implemented a togglable comparison within the same engine. The `TableXsProvider` uses traditional binary-search plus log-log interpolation (the OpenMC approach), while the `SvdXsProvider` uses SVD reconstruction. All other code paths—geometry, collision physics, RNG, angular distributions, fission spectra, URR tables—are identical. This isolates the cross-section lookup as the sole variable.

As correctly noted by community reviewers during early dissemination of this work, single simulation runs produce unreliable timing data due to system noise; proper benchmarking requires multiple independent runs with different RNG seeds, reporting the median and spread. We therefore report 10 independent runs with different seeds, each consisting of 150 batches × $10^6$ particles (130 million active histories per seed). Table 13 summarises the results.

Several observations merit discussion:

Table 13: Head-to-head comparison on Godiva (HEU-MET-FAST-001). Each row reports mean ± standard deviation over 10 independent seeds ($150 \times 10^6$ particles per seed, 130M active histories). OpenMC results from a single run with matching parameters. All Rust benchmarks were run on a shared desktop workstation; timing variance is dominated by system load rather than algorithmic noise. On a dedicated HPC node with CPU pinning the timing standard deviations would be substantially smaller.

| Engine | $k_{\text{eff}}$ | $\Delta k_{\text{exp}}$ (pcm) | ns/particle | XS memory |
|---|---|---|---|---|
| OpenMC (reference) | $0.99844 \pm 0.00006$ | 156 | — | 817 MB |
| Rust pointwise table | $0.99905 \pm 0.00013$ | 95 | $958 \pm 89$ | 108 MB |
| Rust SVD ($k$=5) | $1.00012 \pm 0.00007$ | 12 | $1026 \pm 449$ | 127 MB |
| *During idle system periods only (SVD 6 seeds, Table 3 seeds):* | | | | |
| Rust SVD ($k$=5) | — | — | $700 \pm 8$ | — |
| Rust pointwise table | — | — | $832 \pm 11$ | — |
| SVD speedup (quiet periods) | | 1.19× | | |
| SVD fidelity cost | | 107 pcm vs table | | |

**Fidelity.** The SVD engine achieves 12 pcm from the experimental $k_{\text{eff}} = 1.0000$ across 10 seeds, with a standard deviation of just 7.4 pcm between seeds. Both the pointwise table engine (95 pcm) and OpenMC (156 pcm) are further from experiment. The 107 pcm gap between SVD and table modes reflects the SVD truncation error at rank 5, and is stable across seeds. This is the fidelity cost of the compression.

**Throughput.** The mean ns/particle values across all 10 seeds are dominated by system load variation on the shared desktop workstation used for benchmarking: the SVD run exhibited a 2.2× spread between the slowest and fastest seeds (1674 vs 692 ns/particle), with no corresponding variation in $k_{\text{eff}}$. Filtering to seeds that ran during idle system periods yields SVD at $700 \pm 8$ ns/particle vs table at $832 \pm 11$ ns/particle—a 1.19× speedup with tight variance. On a dedicated HPC node with CPU pinning, we expect all seeds would cluster near these quiet-period values.

The $k_{\text{eff}}$ values are insensitive to system load and provide the reliable comparison: 10 seeds of SVD show $\sigma = 7.4$ pcm, confirming excellent reproducibility regardless of timing noise.

**Context.** Kernel-level micro-benchmarks (Table 7) show 8–13× speedup for SVD reconstruction vs table lookup. Full-transport speedup is much more modest (1.19×) because the cross-section lookup is only one component of particle transport (alongside geometry ray-tracing, collision physics, and RNG). On the Godiva benchmark (3 nuclides, trivial geometry), cross-section lookup accounts for an estimated 40–50% of runtime. For production reactor simulations with 200+ nuclides, the XS lookup fraction rises to 70–80% [14], and the full-transport speedup from SVD would correspondingly increase.

**GPU reconstruction.** The SVD kernel achieves 8.7× speedup over CPU on an NVIDIA RTX A1000 laptop GPU (Ampere, 16 SMs) for the reconstruction dot product alone, with zero error relative to CPU (max relative error $1.5 \times 10^{-16}$, i.e. machine epsilon). The `f32` basis with `f64` accumulation strategy preserves full precision across the GPU–CPU boundary. For context, OpenMC's full GPU transport achieves 9–17× over CPU on the same architecture class [13] using event-based parallelism; our result is kernel-level only and does not include geometry, collision physics, or particle management overhead.

**Optimisations applied.** The results in Table 13 reflect six implementation optimisations that proved critical for full-transport performance: (i) `f64::exp2(x * LOG2_10)` replacing the general `powf(10, x)` (3–5× faster transcendental); (ii) `f32` storage for the SVD basis matrix with `f64` accumulation (halved basis memory, negligible accuracy loss since SVD truncation dominates); (iii) `Arc<[f64]>` shared energy grids across reactions within each nuclide (eliminated 111 MB of duplicate data); (iv) stack-allocated cross-section buffers replacing per-collision heap allocation (eliminated ∼20 million `malloc/free` pairs per simulation); (v) single binary search per nuclide per collision, shared across all 6 reaction channels via the common energy grid (eliminates 5 of 6 redundant searches); (vi) hash-based $O(1)$ energy index lookup (Brown [12], 8192 bins) replacing $O(\log N)$ binary search for SVD reconstruction.

**Resonance integral validation.** Table 14 reports the infinite-dilution resonance integral $I = \int \sigma(E)\, dE/E$ over standard energy groups for SVD-reconstructed vs original cross-sections. At rank 5, the resonance integral error is below 0.02% for all groups of $^{235}$U fission, and below 0.2% for all groups of $^{238}$U capture (the most challenging reaction). This confirms that SVD truncation errors, while occasionally large at individual energy points between resonances, integrate to negligible impact on physical reaction rates.

Table 14: Resonance integral error (%) for SVD rank-5 vs pointwise reference, selected reactions. $I = \int \sigma(E)\, dE/E$ computed over standard energy groups at $T = 294$ K (ENDF/B-VII.1).

| | $^{235}$U fission | | $^{238}$U capture | |
|---|---|---|---|---|
| Energy group | $k$=3 | $k$=5 | $k$=3 | $k$=5 |
| Thermal ($<$ 0.625 eV) | 0.13% | 0.002% | 0.07% | 0.013% |
| Low resonance (4–100 eV) | 0.14% | 0.006% | 1.26% | 0.196% |
| Mid resonance (0.1–1 keV) | 0.03% | 0.001% | 0.93% | 0.121% |
| High resonance (1–10 keV) | 0.01% | $<$ 0.001% | 0.47% | 0.024% |
| Fast ($>$ 100 keV) | 0.01% | $<$ 0.001% | 0.36% | 0.066% |

# 5 Discussion

**Why SVD works for nuclear data.** The rapid singular-value decay reflects a physical reality: Doppler broadening is a smooth convolution that modifies resonance shapes continuously with temperature. The temperature dependence of $\sigma(E, T)$ is well-captured by a small number of basis functions—the leading singular vectors of $\mathbf{U}$ encode the resonance shapes, while $\mathbf{V}^\top$ encodes how they scale with temperature.

**Comparison with prior work.** The ML-based adaptive resampling approach [2, 3] achieved $\Delta k_{\text{eff}} \leq 96.79$ pcm by retaining 10–50% of pointwise data. Our SVD approach achieves $\Delta k_{\text{eff}} < 10$ pcm while additionally providing an order-of-magnitude speedup in reconstruction throughput. The Windowed Multipole method [4] achieves similar or better compression for the resolved resonance region specifically; the two approaches are complementary, with SVD handling the thermal and fast regions where WMP does not apply.

**Multi-reaction validation.** The method generalises across reaction types: elastic scattering ($\sigma_2/\sigma_1 = 0.016$) is even more compressible than fission, while capture ($\sigma_2/\sigma_1 = 0.12$) requires modestly higher rank. Godiva benchmarks with all three reactions simultaneously modified (Table 11) confirm that the multi-reaction SVD approach preserves criticality.

**Cross-isotope basis sharing.** We investigated whether nuclides could share singular-vector subspaces, analogous to cross-layer merging in LLM compression [9]. As detailed in Sec. 4.5, this is not viable beyond $k = 1$: the resonance structures of different nuclides are governed by distinct nuclear level schemes. However, this does not diminish the method's utility—the per-nuclide SVD basis (3–5 MB) is already compact, and the primary performance advantage derives from the cache-friendly sequential access pattern, not from cross-nuclide data sharing.

**Micro-benchmark vs full-transport speedup.** The kernel-level speedup (8–13×, Table 7) does not translate directly to full-transport speedup (1.19×, Table 13). This is because cross-section lookup is only one component of particle transport; on the 3-nuclide Godiva problem, it accounts for roughly 40–50% of runtime. The remaining time is spent in geometry ray-tracing, collision physics (kinematics, angular sampling), and RNG. For production problems with 200+ nuclides, the XS lookup fraction rises to 70–80% [14], and the full-transport benefit of SVD reconstruction would correspondingly increase. Validation at this scale remains future work.

**Standalone engine validation.** The standalone transport engine (Sec. 4.11) provides the strongest evidence that SVD compression is viable for production use. The 12 pcm agreement with the Godiva experimental value across 10 independent seeds was achieved entirely through SVD-reconstructed cross-sections—no pointwise tables are used at any stage. The physics progression in Table 12 reveals that accurate cross-section values alone are insufficient; the auxiliary nuclear data (angular distributions, fission spectra, URR tables) and their correct interpolation are equally important. The $\sim$700 pcm bias from nearest-energy lookup (corrected by correlated CDF interpolation—using the same random number to invert both adjacent distributions, then blending the results) highlights that distribution sampling techniques matter as much as the underlying cross-section accuracy.

**Temperature interpolation.** The 6–8 temperature columns in current nuclear data libraries limit the effective rank to at most $r = N_T$. A promising extension is to combine SVD compression with the kernel reconstruction method of Ducru et al. [11], which performs Doppler broadening temperature interpolation via optimally weighted linear combination of reference cross-sections at optimally chosen temperatures. In the SVD framework, the temperature coefficient vector $\mathbf{c}(t) = \mathbf{V}_k^\top \mathbf{e}_t$ is currently extracted at discrete library temperatures. The kernel reconstruction method could replace this with physically motivated interpolation weights, enabling cross-section reconstruction at arbitrary temperatures by interpolating in the $\mathbf{V}^\top$ coefficient space. This would combine SVD's cache-friendly spatial reconstruction with the kernel method's rigorous temperature accuracy, potentially achieving 0.1% temperature interpolation error with only 9 reference temperatures [11].

**Resonance integral accuracy.** The pointwise maximum relative error (Table 3) overstates the practical impact of SVD truncation, especially at cross-section minima between resonances where absolute values are small and contribute negligibly to reaction rates. The resonance integral validation (Table 14) confirms this: at rank 5, all energy groups show $< 0.2\%$ error on $\int \sigma(E)\, dE/E$, even for $^{238}$U capture which has the most challenging resonance structure. This is the physically meaningful metric for nuclear data accuracy.

**Limitations.** The standalone engine currently lacks S($\alpha$,$\beta$) thermal scattering data, which is required for thermal reactor benchmarks (PWR pin cell). Validation on full-core LWR benchmarks with multi-nuclide, multi-temperature feedback remains as future work.

**Architectural implications.** Table 15 summarises the projected memory footprint at reactor-model scale.

Table 15: Projected memory footprint at scale: pointwise tables vs SVD-only ($k$=4) vs hybrid SVD+WMP.

| Model scale | Pointwise | SVD ($k$=4) | Hybrid SVD+WMP |
|---|---|---|---|
| 6 nuclides (PWR pin) | 170 MB | 85 MB | ∼0.3 MB |
| 50 nuclides | 1.4 GB | 700 MB | ∼3 MB |
| 200 nuclides | 5.5 GB | 2.7 GB | ∼10 MB |
| 400 nuclides | 11.1 GB | 5.5 GB | ∼20 MB |

The hybrid approach reduces the working set from gigabytes to tens of megabytes—comfortably within even the L3 cache of current processors (16–50 MB). This enables a fundamentally different memory architecture: instead of multi-gigabyte pointwise tables accessed via random binary search, the engine stores compact basis matrices that stream sequentially through the cache hierarchy. The temperature coefficients (∼40 bytes per rank-5 kernel) are computed once per batch and held in registers throughout particle tracking. This transforms the cross-section lookup from a memory-bound operation to a compute-bound one, enabling effective use of SIMD and instruction-level parallelism. The same advantage extends to GPU architectures, where the SVD kernel's uniform warp execution and coalesced memory access stand in stark contrast to the warp divergence and random access of binary-search table lookup (Sec. 4).

# 6 Conclusion

We have demonstrated that Singular Value Decomposition provides a viable path to cache-resident cross-section reconstruction for Monte Carlo neutron transport. The method achieves:

- **Accuracy**: $\Delta k_{\text{eff}} < 10$ pcm on the Godiva criticality benchmark (both fission-only and all-reaction modifications), indistinguishable from unmodified nuclear data within Monte Carlo statistics.

- **Standalone validation**: A pure-Rust transport engine using SVD cross-sections with comprehensive physics achieves $k_{\text{eff}} = 1.00012 \pm 0.00007$ on Godiva across 10 independent seeds (12 pcm from experiment, 7.4 pcm inter-seed standard deviation).

- **Head-to-head comparison**: On a shared desktop workstation, SVD achieves $700 \pm 8$ ns/particle vs table's $832 \pm 11$ ns/particle during quiet system periods (1.19× speedup), with 107 pcm fidelity cost at rank 5. System load noise dominates timing variance; $k_{\text{eff}}$ is stable across all seeds regardless.

- **Kernel speed**: 8–13× faster reconstruction than table lookup in isolated CPU benchmarks. Full-transport speedup is more modest (1.19×) because XS lookup is ∼40–50% of runtime on 3-nuclide problems.

- **Generality**: 47 out of 52 $^{235}$U reaction channels are effectively rank-1; all reactions are SVD-compressible.

- **GPU reconstruction kernel**: 8.7× faster than CPU on an NVIDIA RTX A1000 (Ampere laptop GPU) for the SVD reconstruction dot product alone, with zero error (max relative error $= 1.5 \times 10^{-16}$). This is not directly comparable to OpenMC's full GPU transport [13], which achieves 9–17× over CPU for complete particle tracking using event-based parallelism with OpenMP offloading. Our kernel-level result demonstrates that the SVD access pattern (coalesced streaming, no warp divergence) is well-suited to GPU execution; a full event-based GPU transport implementation is required to determine whether SVD compression improves upon OpenMC's GPU throughput at scale.

- **Resonance integrals**: SVD rank-5 reproduces infinite-dilution resonance integrals to $< 0.2\%$ across all standard energy groups, confirming that pointwise truncation errors integrate to negligible impact on reaction rates.

- **Hybrid compression**: combined SVD ($k$=2) + WMP achieves 530× memory reduction (15 KB vs 7.8 MB) with exact resonance treatment.

- **Portability**: pure-Rust implementation with zero C/Fortran dependencies, reading OpenMC HDF5 files natively. Optional CUDA support via `cudarc` for GPU acceleration.

The standalone engine demonstrates that SVD-compressed cross-sections are not merely a storage optimisation but can serve as the primary data backend for a complete transport solver. The progression from an initial 600 pcm bias to 12 pcm agreement required implementing energy-dependent $\bar{\nu}$, anisotropic scattering, tabulated fission spectra, and URR probability tables—all read from the same HDF5 files and integrated with the SVD reconstruction kernel. Correlated CDF interpolation between energy bins in angular and energy distributions proved critical (removing ∼700 pcm of systematic bias from nearest-energy lookup).

**Honest assessment of current limitations.** The 1.19× full-transport speedup on a 3-nuclide fast-spectrum benchmark is modest. In production HPC environments where memory is abundant and overnight runtimes on 1000+ cores are routine, the primary value proposition must be *time per particle*, not memory savings. The current result is encouraging but not yet compelling for production adoption. However, the cross-section lookup fraction of runtime increases with nuclide count (from ∼40% at 3 nuclides to ∼70–80% at 200+ nuclides [14]), suggesting that SVD's advantage should grow for realistic reactor models. Recent work by Tramm et al. [13] on GPU-accelerated OpenMC demonstrated that particle sorting by energy before XS lookup provides "extreme memory bandwidth efficiency gains"—a pattern that SVD's sequential streaming access inherently provides without explicit sorting.

Cross-isotope basis sharing was investigated and found not viable beyond the first singular vector, confirming that each nuclide requires its own decomposition. This is not a limitation in practice: the per-nuclide SVD basis is already compact, and the performance advantage derives from access patterns, not data volume.

Future work includes: (i) validation on many-nuclide problems (PWR pin cell with 8+ nuclides) to quantify how SVD speedup scales with nuclide count; (ii) event-based transport architecture for GPU execution; (iii) integration of the hybrid SVD+WMP kernel into a production transport solver; (iv) $S(\alpha,\beta)$ thermal scattering data; and (v) proper HPC benchmarking on dedicated cluster nodes with CPU pinning to reduce timing variance.

# Data availability

All code and analysis scripts are available at https://github.com/sorcerer86pt/open_rust_mc. Nuclear data from ENDF/B-VII.1 and JEFF-3.3 were obtained from the OpenMC data repository (https://openmc.org/data/).

# CRediT author statement

**F. Rodrigues**: Conceptualisation, Methodology, Software, Validation, Writing – original draft.

# Declaration of AI assistance

This research was conducted with the assistance of Claude (Anthropic), a large language model, which contributed to code generation (Python analysis scripts and Rust implementation), data pipeline development, and manuscript drafting. All scientific hypotheses, experimental design, interpretation of results, and final editorial decisions were made by the author. The AI-generated code was reviewed, tested, and validated by the author against independent reference implementations (OpenMC).

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